SEEING CHANGE FROM ORBIT · MEETING MINUTES

Meeting 2: First SDP I Supervisor Meeting

Date and time	Mon 14 Sep 2026, 1:30–2:30 pm
Type	Supervisor
Location	Online (MS Teams)
Attendees	Dr. Panos, Khaled, Zain, Ayesha, Tehsin, Zayed

Agenda

No agenda was sent. Topics covered:

1. Labelling the data
2. Core workflow and work tracks
3. Tides and multi-temporal measurements
4. Expected outcomes and scope
5. Where to start, SDP I vs SDP II
6. Resources, GitHub and documentation
7. Project plan and next meeting

Member updates

Who	Update
Tehsin	Created the team GitHub repository with the project README (9 Sep)

Discussion

1. Labelling the data

- Khaled raised labelling. Start with datasets that are already labelled, not Abu Dhabi ("start with 2 kg, not 200 kg").
- For unlabelled Abu Dhabi images: run several open-source segmenters, fuse their outputs by majority agreement, then attach meaning to the regions (sea, mangrove, concrete).

2. Core workflow and work tracks

- Core of the project: tracking change in satellite images over time.
- Workflow: synchronise images by timestamp → choose the object of interest (mangroves) → define a region of interest (thin strip around the sea boundary) → anchor the bounding box on invariant landmarks (e.g. the Eastern Mangroves hotel) → segment inside it → combine segmenters (ensemble/meta-learner) → compare over time.
- Possible tracks: data synchronisation; segmenters (some general and fine-tuned, some satellite-specific); combining segmenter outputs; landmark detection; history of open UAE satellite data.

3. Tides and multi-temporal measurements

- Dr. Panos: don't rely on a single acquisition; use several synchronised measurements (week, month, season, multi-year) with time-aware fusion.
- Khaled: tide is a major issue because UAE mangroves are shallow. Dr. Panos: this is part of data synchronisation (including moon cycles); the team should design the approach.

4. Expected outcomes and scope

- Khaled asked whether this is research or benchmark-driven. Dr. Panos read the proposal outcomes: Abu Dhabi evaluation set with weak/self-supervised labels; DINOv3/SAM adapted by light tuning of a frozen model; measured performance drop on Abu Dhabi; how much radar + optical fusion recovers.
- Open-ended research: do the groundwork first, then set realistic, measurable objectives.

5. Where to start, SDP I vs SDP II

- Zain asked for starting papers/tools: use the proposal keywords, DINOv3, and a targeted Scopus review. Analogues: Amazon deforestation and wildfire burned-area mapping.
- Ayesha asked about milestones: SDP I = develop tools, apply them to existing datasets, and if feasible a short Abu Dhabi case study; SDP II = the core change/anomaly detection model. Work backwards from the SDP I aim.

6. Resources, GitHub and documentation

- Department HPC cluster (8× H100) may be usable; otherwise Google Colab.
- Everything goes on GitHub; marks depend on each person's visible contributions.
- Keep an agenda for each meeting (sent the day before), team minutes, and an internal folder of ideas and closed investigations.

7. Project plan and next meeting

- Dr. Panos wants a plan within one week: SDP I aim, milestones, deliverables, tasks and subtasks, who does what for the initial tasks. He will review it and it will be revised as needed.

Decisions

- Build skills on labelled datasets before moving to Abu Dhabi data.
- Abu Dhabi labels come from weak / self-supervised methods (segmenter consensus), not exhaustive hand-labelling.
- Segment only a coastal region of interest anchored on invariant landmarks.
- Use multiple synchronised measurements, not single images; tide is handled as part of synchronisation.
- Targeted literature review first (max ~3 weeks), split between segmentation and change/anomaly detection.
- Each member commits their own work to GitHub.
- Next week's meeting moves to 12:15 (one week only).

Action items

Who	What	Due
Team	Project plan: SDP I aim, milestones, deliverables, tasks, initial allocation	Mon 21 Sep
Team	Targeted literature review, split between segmentation and anomaly detection	~3 weeks
Team	Identify environmental datasets and whether access permission is needed	Not set
Team	Investigate compute (HPC cluster, Colab)	Not set
Team	Investigate open-source segmenters, fusing their outputs, and ensemble/meta-learning methods	Not set
Team	Investigate when UAE satellite data became openly available	Not set
Team	Investigate temporal sampling (weekly, monthly, seasonal, multi-year)	Not set
Team	Prepare an agenda; confirm in person or Teams for next meeting	Before 21 Sep
Dr. Panos	Review the plan and give feedback on feasibility and timeline	After submission

Next meeting

Mon 21 Sep 2026, 12:15 (format to be confirmed).